Teddy S. Rauchway

B.S. Mechanical Engineering, Aerospace Minor — Cornell University, Expected May 2028

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I've been making things for as long as I can remember, from homemade fishing lures and Star Wars helmets to model rockets and remote-controlled airplanes. Watching F-16s take off with my grandfather left me wanting to design aircraft of my own, and that ambition still drives me. I love figuring out how things work and turning my own ideas into working hardware: an enclosure now fielding to multiple defense systems, UAVs I've built and flown, and a wing I designed from scratch and tested in CFD.

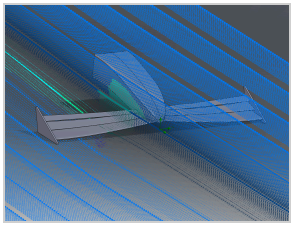
Full project write-ups, CAD files, and build photos/videos available at the site linked above.

SHEET 01 PROJECTS

ICR Enclosure Project

ICR, INC. · SUMMER 2026

Led a clean-sheet redesign of the motorized enclosure that protects fielded defense equipment: a split-lid design on a bolt-together, weld-free frame built from repeated parts, more than 40% lighter than the legacy unit. Ran SolidWorks Motion, CFD (80 mph wind), and FEA, and produced the CAD and drawings for manufacturing. Design complete and fielding to multiple systems.



F1 Front Wing Design & CFD

PERSONAL PROJECT

Self-initiated a deep dive into SolidWorks Premium's Flow Simulation tool, designing a multi-element F1-style front wing from scratch with no class or team behind it. Ran an external flow analysis at 100 mph and measured 347.5 N (≈78 lbf) of downforce by integrating pressure loads across the wing.



UAV Systems

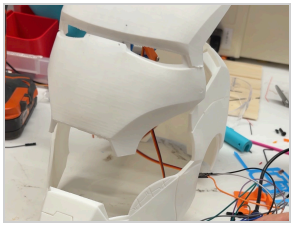
LOCKHEED MARTIN × NORTHFIELD HIGH SCHOOL / PERSONAL

Built and flew UAVs hands-on, not just designed them: as a Northfield High School (Denver, CO) student, worked in tandem with Lockheed Martin engineers to build a LIDAR-mapping hexacopter, running onboard Raspberry Pi/LIDAR mapping software myself. Then designed, wired, and flight-tested a scratch-built fixed-wing autopilot aircraft through several prototypes until it flew clean.

Pulse Power

CORNELL MAKEATHON 2026

Designed, built, and demoed a wearable sleep-stage-detection device in a single 24-hour hackathon, using heart rate and skin temperature sensing to wake the user at the end of a REM cycle. Built with teammates Sani Mohottige and Liam Alcalay.



Iron Man Mark III Helmet

CORNELL MAKER CLUB

Built a functional articulating Iron Man helmet — servo motors coordinated by an Arduino, running on a custom rechargeable battery pack built from reclaimed 18650 cells. Followed a guided Sesame robot build first to learn the basics of servos and circuits.

Vibrating Grape Filtration

CORNELL — CLASS PROJECT

Designed and built the motor-driven vibration linkage for a filtration tray that separates spotted lanternfly contamination from grape juice during harvest, sourcing components from McMaster-Carr. Built with Flynn Basehart, Kathryn Moskal, Emma Harshberger, and Gavin Sneyd.

SHEET 02 SKILLS

CAD & SIMULATION	SolidWorks (CAD, assemblies, drawings, surfacing), SolidWorks Motion, Simulation (FEA) & Flow (CFD), Fusion 360
3D PRINTING	Bambu Lab, Elegoo, Anycubic — daily-driver FDM and resin printers
ELECTRONICS & CODE	Arduino, Raspberry Pi, STM32, Teensy, ESP32; Python; Bluetooth data integration; multimeter & oscilloscope
FABRICATION	CNC mill & lathe (Red Apron certified), sheet metal, soldering, composite surface work, laser cutting, hand & power tools
ELECTROMECHANICAL	BLDC, brushed DC, servo & stepper integration; ESCs; custom gearboxes; sensors; harness fabrication
MANUFACTURING	Company PDM, tolerancing & dimensioning, material selection, assembly documentation

Cornell Maker Club Executive Board (Jan 2026–Present) — overseeing makerspace lab operations, training members on machinery and safety, and staffing hackathons and build nights. **NCAA Division I** sprint hurdler, Cornell Track & Field. **FAA Part 107** Remote Pilot · **SSI Open Water** Diver · English (fluent), Spanish (conversational).